Mathematical Physics I (Fall 2026): Homework #3 Solution

Due Oct. 16, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 4, Problem 7.28

(Note: For Problem 7.28, consider ideal gas. You may want to prove and utilize the findings in Problem 7.27. Note the meaning of the subscripts next to the partial derivatives from Boas Chapter 4, Section 1 in case you did not read it. Prove further that the resulting formula is the familiar $c_p - c_v = nR$ found in e.g., Schroeder or Halliday & Resnick, where n is the number of moles of gas present and R is the gas constant.)

• Replacing $x \to T$, $y \to v$, $z \to p$ and $u \to s$ gives $\left(\frac{\partial s}{\partial T}\right)_p = \left(\frac{\partial s}{\partial T}\right)_v + \left(\frac{\partial s}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_p$, thus $c_p = c_v + T\left(\frac{\partial s}{\partial v}\right)_T \left(\frac{\partial v}{\partial T}\right)_p$. With $\left(\frac{T\partial s}{\partial v}\right)_T = p$ and $\left(\frac{\partial v}{\partial T}\right)_p = \left(\frac{\partial (nRT/p)}{\partial T}\right)_p = \frac{nR}{p}$, you find $c_p - c_v = nR$.

2. Boas Chapter 4, Problem 8.8

(Note: For Problem 8.8, tackle the stated problem in two ways, one with the method in Section 8 and the other with the Lagrange multiplier described in Section 9.)

• The cross section of the gutter is $A = \frac{1}{2} b \sin\theta \{(l-2b) + (l - 2b + 2b\cos\theta)\} = b^2 \sin\theta\cos\theta + (l-2b) b \sin\theta$, where $l = 24$ cm, b is the amount at each side that is bent up, and θ is the angle through which it is bent. Differentiating A with two independent variables θ and b, $\frac{\partial A}{\partial \theta} = 0$ gives $l - 2b + b\cos\theta - \frac{b\sin^2\theta}{\cos\theta} = 0$ while $\frac{\partial A}{\partial b} = 0$ gives $l - 4b + 2b\cos\theta = 0$. Combining the two equations, you reach $b = \frac{l}{3}$ and $\theta = \frac{\pi}{3}$.

• Alternatively, to utilize the Lagrange multiplier, we write $F = A + \lambda l$ from Eq.(9.6) with $A = b\sin\theta(a + b\cos\theta)$ and $l = a + 2b$. Differentiating F with three variables θ, a and b, $\frac{\partial F}{\partial \theta} = 0$ gives $a + b\cos\theta - \frac{b\sin^2\theta}{\cos\theta} = 0$, $\frac{\partial F}{\partial a} = 0$ gives $b\sin\theta + \lambda = 0$, and $\frac{\partial F}{\partial b} = 0$ gives $a + 2b\cos\theta - 2b = 0$. You can easily observe that the three equations here are equivalent to the two equations acquired by the first method.

3. Boas Chapter 4, Problem 12.13

• From Eq.(12.13) with $v(x) = \frac{2}{x}$ and $u(x) = \frac{1}{x}$,

$$\frac{d}{dx}\int_{1/x}^{2/x} \frac{\sin xt}{t} dt = \frac{\sin 2}{2/x} \cdot \left(-\frac{2}{x^2}\right) - \frac{\sin 1}{1/x} \cdot \left(-\frac{1}{x^2}\right) + \int_{1/x}^{2/x} \cos xt dt = 0$$

4. Boas Chapter 5, Problem 3.7

(Note: For Problem 3.7, review the notations introduced in the preceding pages. For (d), review and utilize the two theorems discussed in the current section.)

• (d) From the perpendicular axis theorem for a lamina, $I_z = I_x + I_y = 2M + \frac{9}{2}M = \frac{13}{2}M \rightarrow$ then, from the parallel axis theorem discussed in Problem 3.1, $I_m = I_z - Md^2 = I_z - M(\bar{x}^2+\bar{y}^2) = \frac{13}{2}M - M\left[2^2 + \left(\frac{4}{3}\right)^2\right] = \frac{13}{2}M - \frac{52}{9}M = \frac{13}{18}M$.

5. Boas Chapter 5, Problem 5.1

• From Eq.(5.2) with $\phi(x,y,z) = x - 2y + 5z - 13 = 0$,

$$A = \iint \sec\gamma \, dxdy = \iint dxdy \cdot \frac{|\nabla\phi|}{|\partial\phi/\partial z|} = \int_0^3 dr \int_0^{2\pi} rd\theta \cdot \frac{|(1,-2,5)|}{5}$$
$$= 2\pi \int_0^3 rdr \cdot \frac{\sqrt{30}}{5} = \frac{9\sqrt{30}}{5}\pi.$$

6. Boas Chapter 6, Problem 3.12

• (a) Using Thm.(3.9), $(\mathbf{A}\cdot\mathbf{B})^2 - [(\mathbf{A}\times\mathbf{B})\times\mathbf{B}]\cdot\mathbf{A} = (\mathbf{A}\cdot\mathbf{B})^2 - [-(\mathbf{B}\cdot\mathbf{B})\mathbf{A} + (\mathbf{A}\cdot\mathbf{B})\mathbf{B}]\cdot\mathbf{A} = (\mathbf{A}\cdot\mathbf{B})^2 + A^2B^2 - (\mathbf{A}\cdot\mathbf{B})^2 = A^2B^2$.

• (b-1) Using Eq.(3.3) and Thm.(3.9), $(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) = \mathbf{A}\cdot[\mathbf{B}\times(\mathbf{C}\times\mathbf{D})] = \mathbf{A}\cdot[(\mathbf{B}\cdot\mathbf{D})\mathbf{C} - (\mathbf{B}\cdot\mathbf{C})\mathbf{D}] = (\mathbf{A}\cdot\mathbf{C})(\mathbf{B}\cdot\mathbf{D}) - (\mathbf{A}\cdot\mathbf{D})(\mathbf{B}\cdot\mathbf{C})$. You may prove the identity in a brute force approach using Eq.(3.2), too.

• (b-2) Alternatively, writing in tensor notation and then using Eqs. (5.8) and (5.11) of Boas Chapter 10, $(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) = (\mathbf{A}\times\mathbf{B})_i(\mathbf{C}\times\mathbf{D})_i = \epsilon_{ijk}A_jB_k \cdot \epsilon_{imn}C_mD_n = (\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km})A_jB_kC_mD_n = A_jB_kC_jD_k - A_jB_kC_kD_j = (\mathbf{A}\cdot\mathbf{C})(\mathbf{B}\cdot\mathbf{D}) - (\mathbf{A}\cdot\mathbf{D})(\mathbf{B}\cdot\mathbf{C})$.

7. Boas Chapter 6, Problem 6.16

• Using Eq.(9.6) of Chapter 4 and Eq.(6.2) of Chapter 6, we write $F = \frac{d\phi}{ds} + \lambda(a^2+b^2+c^2) = \left(\frac{\partial\phi}{\partial x}a + \frac{\partial\phi}{\partial y}b + \frac{\partial\phi}{\partial z}c\right) + \lambda(a^2+b^2+c^2)$. Then, $\frac{\partial F}{\partial a} = \frac{\partial\phi}{\partial x} + 2\lambda a = 0$, $\frac{\partial F}{\partial b} = \frac{\partial\phi}{\partial y} + 2\lambda b = 0$, $\frac{\partial F}{\partial c} = \frac{\partial\phi}{\partial z} + 2\lambda c = 0$, and $a^2+b^2+c^2 = 1$. Combined, $\lambda = \pm\frac{|\nabla\phi|}{2}$ and $\mathbf{u} = (a,b,c) = -\frac{\nabla\phi}{2\lambda} = \pm\frac{\nabla\phi}{|\nabla\phi|}$. With Eq.(6.4), you get the extremum values of $\frac{d\phi}{ds}$ as $\nabla\phi\cdot\mathbf{u} = \pm\frac{(\nabla\phi)^2}{|\nabla\phi|} = \pm|\nabla\phi|$.

8. Boas Chapter 6, Problem 9.4

• By setting $P = e^x\cos y$ and $Q = -e^x\sin y$ and applying Eq.(9.7) of Chapter 6,

$$\iint_{BADB}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dxdy = 0 = \oint_{BADB} Pdx + Qdy$$
$$= \left(\int_{BA} Pdx + Qdy\right) + \left(\int_{ADB} Pdx + Qdy\right)$$
$$= \left(\int_{(-\ln 2,0)}^{(\ln 2,0)} Pdx + Qdy\right) + \left(\int_C Pdx + Qdy\right)$$

$$= \left(\int_{-\ln 2}^{\ln 2} e^x dx \right) + \left(\int_C P dx + Q dy \right)$$
$$= \frac{3}{2} + \left(\int_C P dx + Q dy \right).$$

9. A localized, axisymmetric charge density distribution is given by $\rho(r, \theta) = \rho_0 \frac{\cos\theta}{r}$ within the region $a \leq r \leq b$ centered on the origin, where θ is measured from the z-axis.

(a) Calculate the net enclosed charge $q = \int \rho(r, \theta) dV$ over the region.

(b) Compute the total z-component of the dipole moment $p_z = \int (r \cos\theta) \rho(r, \theta) dV$.

Now suppose we are interested in the electric field far from a localized charge distribution. In this limit, the axisymmetric electrostatic potential is written as:

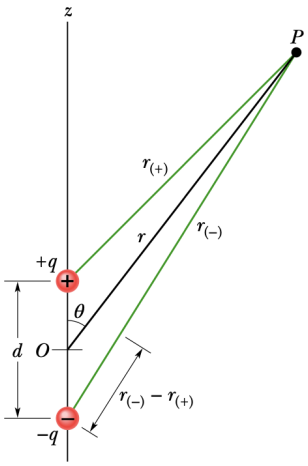
$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} + \frac{p \cos\theta}{r^2} \right)$$

where q is the monopole charge and p is the dipole moment.

(c) Compute the electric field components in polar coordinates, (E_r, E_θ).

(d) For a pure dipole ($q = 0$), find all angles $\theta \in [0, \pi]$ where the radial component E_r vanishes.

(e) Using the multivariable chain rule, evaluate $\frac{dV}{dt}$ along the path $(r(t), \theta(t)) = (r_0 e^{\alpha t}, \omega t)$.



[Adapted from Halliday & Resnick, Chapter 24.4]

(Note: To learn more about the electric dipole, you may want to review the freshman physics textbooks such as Halliday & Resnick, or the textbooks in electromagnetism such as Griffiths.)

- (b) $p_z = \rho_0 \int_0^{2\pi} d\phi \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta \int_a^b r^2 \, dr = \frac{4\pi\rho_0}{9}(b^3 - a^3)$.
- (e) Applying the chain rule $\frac{dV}{dt} = \frac{\partial V}{\partial r}\frac{dr}{dt} + \frac{\partial V}{\partial \theta}\frac{d\theta}{dt}$,

$$\frac{dV}{dt} = \left(-\frac{1}{4\pi\epsilon_0}\frac{q}{r^2} - \frac{1}{4\pi\epsilon_0}\frac{2p\cos\theta}{r^3} \right)(\alpha r) + \left(-\frac{1}{4\pi\epsilon_0}\frac{p\sin\theta}{r^2} \right)\omega = -\frac{1}{4\pi\epsilon_0}\left[\frac{q\alpha}{r} + \frac{p}{r^2}(2\alpha\cos\theta + \omega\sin\theta) \right].$$

10. (a) Review Boas Chapter 6, Example 7.2, thus prove Eq.(7.6) or Eq.(f) in p.339. You are also asked to prove the identity by explicitly working with the components, i.e.,

$$\nabla \cdot (\phi \mathbf{V}) = \frac{\partial(\phi V_x)}{\partial x} + \frac{\partial(\phi V_y)}{\partial y} + \frac{\partial(\phi V_z)}{\partial z} = ...$$

(b) Review how Gauss's law $\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$ is acquired from Coulomb's law in Boas Chapter 6, Section 10, where $\mathbf{E}$ is the electric field, ρ is the volume charge density, and ϵ_0 is the permittivity of free space.

(c) Using (a) and (b), show that the energy of a continuous charge distribution is given by

$$W_{\mathrm{e}} = \frac{1}{2}\int \rho V d\tau = \frac{\epsilon_0}{2}\int |\mathbf{E}|^2 d\tau$$

for integration over all space, where V is the electric potential. You are first asked to briefly discuss how one arrived at the term $W_{\mathrm{e}} = \frac{1}{2}\int \rho V d\tau$, and then explicitly prove the second equality. You may assume that V vanishes at large distance r at least as fast as r^{-1}.

(d) Now, prove Eq.(h) in p.339. You may prove the identity by explicitly working with the components, i.e.,

$$\nabla \cdot (\mathbf{U} \times \mathbf{V}) = \frac{\partial(\mathbf{U} \times \mathbf{V})_x}{\partial x} + \frac{\partial(\mathbf{U} \times \mathbf{V})_y}{\partial y} + \frac{\partial(\mathbf{U} \times \mathbf{V})_z}{\partial z} = ...$$

(e) Review how Ampere's law $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ is acquired from the Ampere's circuital law in Boas Chapter 6, Section 11, where $\mathbf{B}$ is the magnetic field, $\mathbf{J}$ is the current density, and μ_0 is the permeability of free space.

(f) Using (d) and (e), show that the energy stored in magnetic fields is given by

$$W_{\mathrm{m}} = \frac{1}{2}\int (\mathbf{A} \cdot \mathbf{J}) d\tau = \frac{1}{2\mu_0}\int |\mathbf{B}|^2 d\tau$$

for integration over all space, where $\mathbf{A}$ is the magnetic vector potential. You are first asked to briefly discuss how one arrived at the term $W_{\mathrm{m}} = \frac{1}{2}\int (\mathbf{A} \cdot \mathbf{J}) d\tau$, and then explicitly prove the second equality. You may assume that $\mathbf{A}$ vanishes at large distance r at least as fast as r^{-1}.

(g) Finally, using Maxwell's equations in vacuum with sources,

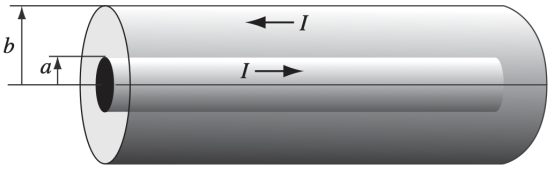
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t},$$

and the identity found in (d), derive Poynting's theorem:

$$\mathbf{E} \cdot \mathbf{J} = -\frac{\partial u}{\partial t} - \nabla \cdot \mathbf{S},$$

where $u = \frac{1}{2}\left(\epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0}|\mathbf{B}|^2\right)$ is the electromagnetic energy density and $\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B})$ is the Poynting vector. Briefly state the physical interpretation of the theorem.

(Note: The exercises here should sound familiar to most of you as you have begun to study and explore electromagnetism. If not, you may want to briefly review the classic textbooks in electromagnetism such as Griffiths (Chapters 2, 7, and 8) or Jackson. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be



[Adapted from Griffiths, Chapter 7.2]

your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)

• (c) One can think of $\frac{1}{2}\int \rho V d\tau$ as the continuum approximation of $W_{\rm e} = \frac{1}{2}\sum_i q_i V(\mathbf{r}_i)$ (see Section 2.4.2 of Griffiths for more information). Then, one gets

$$\begin{aligned} W_{\rm e} &= \frac{1}{2}\int \rho V d\tau = \frac{\epsilon_0}{2}\int V(\nabla\cdot\mathbf{E})d\tau = \frac{\epsilon_0}{2}\int \left[-\mathbf{E}\cdot(\nabla V) + \nabla\cdot(V\mathbf{E})\right] d\tau \\ &= \frac{\epsilon_0}{2}\left(\int |\mathbf{E}|^2 d\tau + \oint_{\partial\tau} (V\mathbf{E})\cdot\mathbf{n} d\sigma\right), \end{aligned}$$

where the second term vanishes as the integration is carried out over all space — since it behaves like $(r^{-1}r^{-2})\cdot r^2 = r^{-1}$ which tends to 0 as r approaches ∞.

• (f) One can think of $\frac{1}{2}\int(\mathbf{A}\cdot\mathbf{J})d\tau$ as the generalization of a energy stored in a current-carrying loop, $\frac{1}{2}\oint(\mathbf{A}\cdot\mathbf{I})dl$, which itself came from $W_{\rm m} = \frac{1}{2}LI^2 = \frac{1}{2}I(\int \mathbf{B}\cdot d\mathbf{a}) = \frac{1}{2}I(\oint \mathbf{A}\cdot d\mathbf{l}) = \frac{1}{2}\oint(\mathbf{A}\cdot\mathbf{I})dl$ (see Section 7.2.4 of Griffiths for more information). Then, one gets

$$\begin{aligned} W_{\rm m} &= \frac{1}{2}\int(\mathbf{A}\cdot\mathbf{J})d\tau = \frac{1}{2\mu_0}\int \mathbf{A}\cdot(\nabla\times\mathbf{B})d\tau = \frac{1}{2\mu_0}\int \left[\mathbf{B}\cdot(\nabla\times\mathbf{A}) - \nabla\cdot(\mathbf{A}\times\mathbf{B})\right] d\tau \\ &= \frac{1}{2\mu_0}\left(\int |\mathbf{B}|^2 d\tau - \oint_{\partial\tau} (\mathbf{A}\times\mathbf{B})\cdot\mathbf{n} d\sigma\right), \end{aligned}$$

where the second term vanishes as the integration is carried out over all space.

• (g) Starting from the Ampere-Maxwell's law,

$$\begin{aligned} \mu_0\mathbf{J} = \nabla\times\mathbf{B} - \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} \quad \rightarrow \quad \mathbf{E}\cdot\mathbf{J} &= \frac{1}{\mu_0}\mathbf{E}\cdot(\nabla\times\mathbf{B}) - \epsilon_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} \\ &= \frac{1}{\mu_0}\left[\mathbf{B}\cdot(\nabla\times\mathbf{E}) - \nabla\cdot(\mathbf{E}\times\mathbf{B})\right] - \epsilon_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} \\ &= -\frac{1}{\mu_0}\mathbf{B}\cdot\frac{\partial\mathbf{B}}{\partial t} - \nabla\cdot\mathbf{S} - \epsilon_0\mathbf{E}\cdot\frac{\partial\mathbf{E}}{\partial t} \\ &= -\frac{\partial u}{\partial t} - \nabla\cdot\mathbf{S}, \end{aligned}$$

with u and $\mathbf{S}$ defined in the problem statement (see Section 8.1.2 of Griffiths for more information).